

Competition Fragility in Intent-Based DeFi Exchanges: Measuring Visible Replacement Depth and Designing Shadow-Reserve Benchmarks

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Abstract

Intent-based decentralized exchanges replace explicit path selection with competition among off-chain solvers. Execution quality therefore depends not only on registered solver count, but also on visible replacement depth: whether the winning solver faces close alternatives in the auction where an intent is executed. We study CoW Protocol after the CIP-67 transition using 802,816 mainnet solver competitions from June 2025 through January 2026. A key measurement correction is solver-level: in the post-CIP-67 single-winner setting, the relevant reference score is the best non-winning solver score, not the second-highest solution overall.

We find that solver competition is improving but remains locally thin. Overall, 25.3% of auctions have fragility above 0.5; among small-notional auctions, the share is 63.1%. Fragile share falls descriptively from 28.1% in June to 20.0% in January, but concentration persists: among cells with at least 50 auctions, 70.5% exceed HHI 0.25. The central empirical finding is a blue-chip paradox: blue-chip cells have lower long-run HHI and more active solvers than long-tail cells, yet blue-chip auctions exhibit higher one-shot fragility. Thus, long-run solver diversity and auction-level replacement depth are distinct market-health dimensions.

Dollar outcomes are interpreted as score-implied accounting-gap proxies, not welfare-loss estimates. On the USD-covered, volume-filtered sample, the visible replacement accounting gap is \$5.13M over \$54.69B of notional, or 0.94 bps. Finally, we show that protocol safeguards should distinguish absolute quality failures from relative winner dominance. The quality-floor trigger flags only 14.4% of the highest-fragility decile. By contrast, simple depth-threshold triggers flag the full decile and overlap with the quality floor in only about 15–16% of triggered auctions. Intent protocols therefore need both absolute quality floors and relative depth monitors.

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1 Introduction

Intent-based exchanges [5] change the locus of execution in decentralized finance. Instead of sending a user through a public router, the protocol exposes the user’s intent to specialized off-chain solvers. In principle, this architecture should improve execution quality: solvers compete over routing, private inventory, RFQ liquidity, internalization opportunities, gas strategy, and cross-order matching. In practice, the execution-quality promise depends on a local market-design question. At the time an intent is auctioned, does the winner face a close replacement, or is the trade effectively executed by a single visible solver?

This paper studies that question in CoW Protocol [7] after its CIP-67 transition [6, 8]. We deliberately frame the object of study as *visible replacement depth*, not full market power. Our data reveal the scores submitted by visible solvers in the post-CIP-67 solver-competition records [6]; they do not reveal all private liquidity, private communications, or the globally optimal counterfactual route. The appropriate measurement target is therefore the depth

that can be audited from the solver competition itself. We ask whether the winning solver is disciplined by a best non-winning solver, whether market cells have persistent concentration, whether visible replacement gaps have material economic scale, and what kind of protocol-side safeguard should respond to different local failures.

The empirical setting has a useful accounting property. In the post-CIP-67 sample, auctions have a single winning solver. For such auctions, the CoW reference score for reward accounting is the highest score among non-winning solvers. We therefore define

$$\text{Fragility}_a = 1 - \frac{\text{Score}_{2,a}^{\text{other}}}{\text{Score}_{1,a}}, \quad \text{Score}_{2,a}^{\text{other}} = \max_{s \neq s^*} \text{Score}_{s,a}.$$

This is not a heuristic runner-up proxy: it is the visible reference score in the observed single-winner setting. Nevertheless, we explicitly validate the parser because a solver may submit multiple candidate solutions, and the second-highest solution overall need not come from a non-winning solver.

Our dataset contains 802,816 mainnet solver competitions from June 2025 through January 2026. The results show both improvement and persistence. Overall, 25.3% of auctions have fragility above 0.5; among small-notional auctions, the share is 63.1%. Mean competitive solver count rises over time, from 2.41 in June to 2.74 in January, and the monthly fragile share falls from 28.1% to 20.0%. A cell-month panel confirms this descriptive market-deepening pattern: fragile share falls by about 0.93 percentage points per month, and higher mean effective solver count is associated with lower fragile share. Yet the market remains locally concentrated. Raw micro-cells are sparse, but support-filtered concentration measures still show high concentration, especially in long-tail cells.

We also quantify economic scale, while avoiding scope overclaims. Dollar replacement-gap proxies mechanically share score components with fragility, so we interpret them as scale translations rather than causal estimates. We use “accounting-gap proxy” for the score-implied difference between the winner and the best visible non-winning solver; it is not an estimate of solver profit, user surplus loss, or total extractable value. On the USD-covered sane-volume sample, this accounting gap totals \$5.13M over \$54.69B of covered notional, or 0.94 bps. This gives the measurement exercise economic meaning without claiming to identify total counterfactual execution loss.

Finally, we design and evaluate a shadow-reserve benchmark. The key design lesson is that there are two distinct failure modes. A quality-floor reserve asks whether the winning score is weak relative to a historical cell benchmark. A depth trigger asks whether the winner lacks a close visible replacement in the same auction. Our depth-targeted diagnostic shows that the two triggers are not substitutes: the baseline quality floor flags only 14.4% of the highest-fragility decile, while simple depth triggers flag 100% of that decile and overlap with the quality floor in only about 15–16% of triggered auctions. Thus the reserve is best understood as one component of a two-trigger monitoring framework, not as a complete fragility intervention.

Our contributions are threefold. First, we operationalize visible replacement depth for intent-based solver markets and validate the best-non-winner reference-score construction. Second, we provide a large-scale empirical measurement of auction-level depth, local concentration, economic scale, missingness, persistence, blue-chip/long-tail heterogeneity, and alternative hypotheses. Third, we distinguish absolute quality floors from relative depth monitors. We characterize the manipulation-resistance trade-offs of endogenous benchmarks and empirically show that many local cells require parent-cell inheritance or audit-only reserves rather than independent hard benchmarks.

2 Background, Model, and Related Work

In an intent-based exchange, a user signs an intent rather than a path [20, 1, 2]. Solvers then submit candidate settlements. A settlement can use public AMMs [3], private inventory, RFQs, batch coincidence of wants, or other execution sources. The protocol validates solutions, assigns scores, and selects a winning solution. This differs from a conventional router: routing intelligence is delegated to a competitive solver market.

We model an auction as $a = (O_a, S_a, B_a)$, where O_a is the set of orders, S_a is the set of solvers, and B_a is the set of submitted solutions. A solution $b_{s,a}$ has score $Score_{s,a}$ after feasibility checks. Let s^* denote the winning solver and $Score_{1,a} = Score_{s^*,a}$. In our post-CIP-67 sample, each auction has a single winning solver. The reference score is the best non-winning score,

$$RefScore_a = Score_{2,a}^{other} = \max_{s \neq s^*} Score_{s,a},$$

with value zero if no non-winning solver submits a positive valid score. The visible fragility of an auction is $1 - RefScore_a / Score_{1,a}$. A value near zero means that the winner is closely replaceable by another visible solver; a value near one means that the visible replacement is weak.

This paper connects three literatures. First, intent-based exchanges, cross-chain intent standards, and order-flow auctions study how user orders are routed through solvers, fillers, and RFQ relationships [2, 4, 8, 12, 23]. Our focus is not routing optimality in general but the local thickness of the solver competition that disciplines execution. Second, MEV and PBS measurement work documents concentration among searchers, builders, and arbitrageurs [9, 13, 14, 19, 22]. We bring a similar market-structure lens to the solver layer. Third, auction design and robust statistics study reserve prices, endogenous benchmarks, and manipulation under repeated observations [11, 15]. Our shadow-reserve construction adapts those ideas to a setting in which a benchmark is generated from historical solver competitions and may itself be strategic.

We avoid three overclaims. We do not infer illegal collusion. We do not identify total counterfactual execution loss relative to an unobserved global optimum. We do not claim that the shadow reserve is a complete deployed fallback mechanism. The object is narrower but auditable: visible replacement depth and benchmark-based accounting-gap compression inside solver competitions.

Why visible replacement depth is different from global liquidity.

A router benchmark answers whether a single execution path was good relative to available public liquidity. Solver competition asks a different question: whether the auction itself disciplines the winner. A solver may win because it has access to a private RFQ, inventory, or routing path that a public router cannot reproduce. Such an advantage may improve user execution in absolute terms while still leaving the user dependent on a single solver in that auction. Conversely, a market cell may have many long-run winners and low HHI but still exhibit a one-shot gap between the best and second-best solver. This is why the paper keeps relative fragility, notional-normalized gap, effective solver count, and cell-level HHI as separate objects.

133 A minimal information-advantage model.

134 The blue-chip paradox in Section 4 is not a contradiction. A simple score decomposition
135 explains why long-run diversity can coexist with one-shot replacement failure. Write solver
136 i 's score in auction a as

$$137 \quad \text{Score}_{i,a} = P_{i,a} + I_{i,a}\Delta_{i,a},$$

138 where $P_{i,a}$ is the public component from routing, AMM access, gas strategy, and standard
139 optimization, while $I_{i,a}\Delta_{i,a}$ is a solver-specific private opportunity such as RFQ access,
140 inventory, internalization, or a temporary route unavailable to the other visible solvers.
141 Adding more solvers can rotate winners over time and reduce long-run HHI. But if liquid
142 blue-chip markets also create more frequent or larger solver-specific opportunities, the realized
143 winner in a particular auction can still be far ahead of the best non-winning solver. The
144 model is not meant to estimate primitives or characterize equilibrium behavior. Its purpose
145 is narrower: it shows why lower cell-level concentration need not imply lower auction-level
146 fragility. Long-run winner diversity and one-shot replacement depth are different market-
147 health dimensions.

148 Connection to order-flow auctions.

149 In traditional order-flow auctions, competition among market makers is valuable only to the
150 extent that losing quotes are close enough to constrain the winner. A nominally open auction
151 can still be thin if only one bidder has the inventory, infrastructure, or information to quote
152 competitively. Intent-based exchanges create a crypto-native analogue of this problem. The
153 solver market is programmable and auditable, but its local depth must still be measured at
154 the auction level.

155 Connection to MEV and PBS concentration.

156 Prior MEV and proposer-builder separation work shows that technical intermediation layers
157 often concentrate around entities with private order flow, better latency, exclusive relation-
158 ships, or inventory advantages. Solver markets are a related but distinct intermediation layer.
159 They face more explicit scoring rules and protocol-side accounting, which lets us compute
160 visible replacement depth directly. The measurement problem is therefore more structured
161 than generic MEV extraction, even though the economic forces are similar.

162 Connection to reserve prices and robust benchmarks.

163 Reserves in repeated auctions can protect a seller or buyer from weak competition, but
164 endogenous reserves are themselves manipulable. A benchmark based on historical solver
165 scores must therefore be delayed, filtered, robustified, and rate-limited. The shadow-reserve
166 mechanism in this paper should be read in that lineage: it is a robust benchmark design for a
167 transparent repeated auction, not a claim that a single reserve price solves all solver-market
168 failures.

169 3 Data and Measurement

170 The main sample contains 802,816 CoW Protocol mainnet solver competitions from June
171 1, 2025 through January 31, 2026. We parse solver-competition JSON records in the CIP-
172 67 era [8, 6]. We retain valid, non-filtered, positive-score solutions. The parser records

■ **Table 1** Sample construction and denominators used in different analyses.

Sample	N	Filter / reason
Main auction sample	802,816	parsed post-CIP-67 auctions
Score-to-USD sample	680,839	score, fragility, controls non-null
USD notional sample	493,653	score_gap_bps non-null
Sane-volume denominator	318,469	volume_usd in [\$1,\$10M]
Replay evaluation sample	642,252	first 20% warm-up excluded

■ **Algorithm 1** Reference-score construction

Require: Solver-competition record with submitted solutions, solver identities, validity flags, and scores

- 1: Retain valid, non-filtered, positive-score solutions.
- 2: Identify the winning solver s^* and winning score $Score_{1,a}$.
- 3: Remove all candidate solutions submitted by s^* .
- 4: **if** at least one positive valid non-winning solver score remains **then**
- 5: $Score_{2,a}^{other} \leftarrow \max_{s \neq s^*} Score_{s,a}$.
- 6: **else**
- 7: $Score_{2,a}^{other} \leftarrow 0$.
- 8: **end if**
- 9: $Fragility_a \leftarrow 1 - Score_{2,a}^{other} / Score_{1,a}$.

173 auction time, token pair or category, winning solver, valid submissions, winning score, best
174 non-winning solver score, effective solver counts, and market-cell labels.

175 Table 1 reconciles the denominators used throughout the paper. Full-sample results such
176 as fragility, effective solver counts, and HHI use all parsed auctions. USD-denominated
177 quantities require price coverage and therefore use smaller denominators. The difference
178 between 802,816 auctions, 680,839 score-to-USD observations, 493,653 notional-covered
179 observations, and 318,469 sane-volume observations is therefore intentional, not a data
180 inconsistency.

181 Reference-score construction and parser audit.

182 Our unit of observation is a solver-competition record after applying the protocol-side validity
183 and filtering flags. We group candidate solutions by solver identity, not by solution identity.
184 The retained post-CIP-67 records have exactly one winning solver, so the reward-accounting
185 reference score is the best positive score submitted by a non-winning solver. This definition is
186 solver-level: a winning solver may submit multiple candidate solutions, so the second-highest
187 solution overall can still come from the winner. The parser therefore removes all candidate
188 solutions submitted by the winning solver and computes

$$189 \quad Score_{2,a}^{other} = \max_{s \neq s^*} Score_{s,a},$$

190 with value zero if no remaining positive valid score exists. Table 2 reports a full-sample
191 winner-cardinality and parser audit. The audit confirms that all retained records have a
192 single winning solver, that 15.2% have no positive non-winning score and therefore receive
193 $Score_{2,a}^{other} = 0$, and that the old solution-level parser would differ from the best-non-winning-
194 solver construction in 11.8% of auctions. All main results in this version use the corrected
195 $Score_{2,a}^{other}$ construction.

■ **Table 2** Full-sample winner-cardinality and parser audit ($N = 802,816$). The paper uses $Score_{2,a}^{other} = \max_{s \neq s^*} Score_{s,a}$, the best score after removing all candidate solutions submitted by the winning solver. The audit shows why solution-level runner-up parsing is insufficient.

Check	Value
Main sample auctions	802,816
Records with exactly one winning solver	802,816
Records with multiple winning solvers retained	0
Auctions with ≥ 1 non-winning solver	680,839
No positive non-winning solver; $Score_{2,a}^{other} = 0$	15.2%
$Score_2^{solution} = Score_{2,a}^{other}$	88.2%
Affected by winner multi-solution submissions	11.8%
Affected auctions (n)	95,069
Median fragility change among affected	0.051
Parser invariant: $Score_{2,a}^{other} \leq Score_{1,a}$	100%
Current parser removes winning solver before reference score	Yes
Direction of bias in old parser	Underestimates fragility

196 Metrics.

197 We use four complementary measures. The relative replacement measure is $Fragility_a =$
 198 $1 - Score_{2,a}^{other} / Score_{1,a}$. The effective solver count is

$$199 \quad N_a^{comp}(\epsilon) = |\{s : Score_{s,a} \geq (1 - \epsilon)Score_{1,a}\}|,$$

200 with $\epsilon = 5\%$ in the main tables. Local concentration is cell-level HHI over winning share.
 201 The USD-normalized gap is

$$202 \quad Gap_a^{bps} = 10^4 \frac{(Score_{1,a} - Score_{2,a}^{other}) / 10^{18} \cdot ETHUSD_a}{VolumeUSD_a},$$

203 reported only where USD notional is available.

204 USD coverage is incomplete but reported explicitly. Appendix Table 13 shows that
 205 covered auctions are not more fragile on the median than uncovered auctions; the covered
 206 sample has more blue-chip auctions and higher effective solver counts. We therefore compute
 207 all non-USD core measurements on the full sample and interpret dollar-scale results as
 208 coverage-conditioned.

209 Market cells.

210 A market cell groups auctions by the economic dimensions along which solver capabilities
 211 plausibly differ. Token pair or category matters because routing graphs, RFQ relationships,
 212 and inventory constraints are token-specific. Size matters because a solver that can quote
 213 small orders may not have balance-sheet capacity for larger orders. Time matters because
 214 liquidity and solver participation change. Chain matters in multi-chain deployments because
 215 settlement cost, liquidity, and solver infrastructure differ. Our baseline cells are deliberately
 216 local; we then recompute concentration under coarser definitions and minimum-support
 217 thresholds so that the results do not rest on a single granularity choice.

218 Score units and USD enrichment.

219 CoW scores are denominated in protocol accounting units that we convert to ETH and then
 220 to USD for scale interpretation. This conversion is used only for economic-scale tables and
 221 Gap_a^{bps} . The core visible-depth measure, $Fragility$, is dimensionless and does not depend

on USD coverage. For notional normalization, we require a sell-token price and a positive sane-volume denominator. This is why the USD notional sample is smaller than the full sample. We explicitly separate full-sample conclusions from USD-conditioned conclusions in Tables 1 and 13.

Filtering and winsorization.

We exclude invalid or filtered-out solver submissions and zero-score solutions. For economic-scale aggregation, we winsorize the dollar replacement-gap proxy at the first and ninety-ninth percentiles to prevent a small number of extreme score conversions from dominating the total. The bps denominator further requires $volume_usd \in [1, 10^7]$ in the economic-scale calculation. The regression and decile tables use the relevant non-missing outcome variables without using them to define the full-sample fragility statistics.

What the data cannot show.

The data do not reveal private negotiations, solver inventory, or all feasible external fallback routes. A high-fragility auction may still be better for users than a public route if the winner has valuable private liquidity. Conversely, low fragility does not prove an auction is globally optimal. We therefore interpret fragility as visible replacement depth in the solver auction, and we use external language such as “user execution harm” only cautiously.

4 Empirical Evidence: Depth, Concentration, and Scale

Table 3 summarizes the eight-month measurement sample, and Figure 1 visualizes the segment-level contrast. Across 802,816 solver competitions, 25.3% of auctions have visible fragility above 0.5 and 20.2% exceed 0.75. The median auction is not fragile in this threshold sense: median fragility is 0.086 and the mean number of competitive solvers is 2.72. The important pattern is therefore not that every auction is broken, but that the tail of weak replacement depth is thick and operationally relevant. Small-notional auctions are the clearest example. They cross the 0.5 fragility threshold in 63.1% of cases and have mean competitive-solver count only 1.13.

Local concentration persists under support filtering.

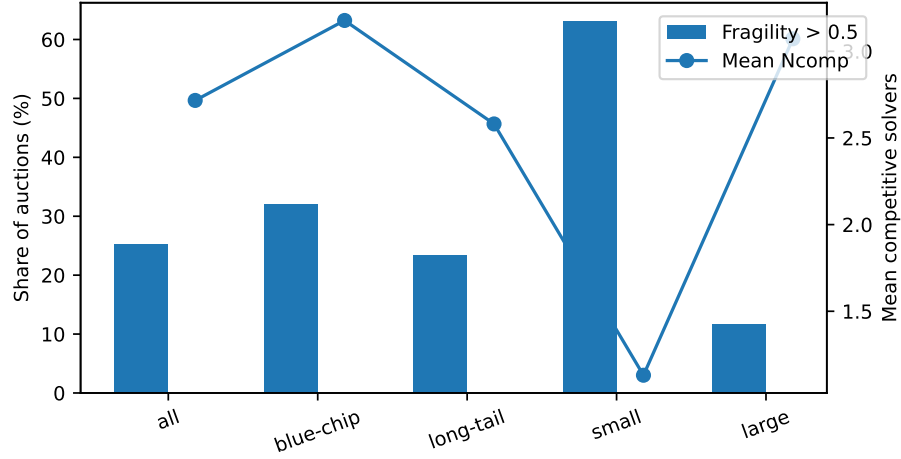
We use HHI as a descriptive local-concentration statistic, following standard antitrust and auction-market practice [21]. The raw cell distribution is sparse, so the all-cell median HHI of 1.0 should not be interpreted as every economically relevant market being a monopoly. The more important result is robustness. In Table 4, after requiring at least 50 auctions per cell, the median HHI remains 0.332 and 70.5% of cells still exceed HHI 0.25. Thus micro-cell sparsity amplifies raw concentration statistics, but it does not explain them away. Appendix Table 14 reports the raw segment-level HHI summary; the main text emphasizes the support-filtered result because it is less vulnerable to sparse-cell artifacts.

The blue-chip paradox.

The most revealing heterogeneity is that long-run diversity and one-shot replacement depth diverge. Blue-chip cells are much less concentrated than long-tail cells: median HHI is 0.343 rather than 1.0, and the median blue-chip cell has seven active solvers. Yet blue-chip auctions have a higher fragile-auction share than long-tail auctions, 32.1% versus 23.3%. Table 5

■ **Table 3** Visible replacement depth by market segment (Jun. 2025–Jan. 2026).

Segment	N	Frag > 0.5	Frag > 0.75	Med. Frag.	Mean N^{comp}	Med. Gap (bps)	USD cov.
all	802,816	25.3%	20.2%	0.086	2.72	1.21	61.5%
blue-chip	182,417	32.1%	27.3%	0.092	3.18	0.29	96.9%
long-tail	620,399	23.3%	18.2%	0.085	2.58	2.56	51.1%
small	200,704	63.1%	54.7%	0.938	1.13	1.29	62.6%
large	200,704	11.6%	7.8%	0.038	3.07	1.28	58.3%



■ **Figure 1** Visible replacement depth by segment. Bars show the share of auctions with fragility above 0.5; the line reports mean effective solver count.

summarizes this contrast. The interpretation is not that blue-chip markets are worse in every sense. Instead, blue-chip markets have more solver diversity over time, but individual auctions can still be won by a solver with a transient RFQ, inventory, gas, routing, or internalization advantage that leaves the best visible non-winning solver far behind. Long-tail markets have the opposite pathology: they are persistently concentrated at the cell level and have larger notional-normalized gaps, even though their relative fragile share is lower. This distinction is the central measurement lesson: HHI and fragility are different layers of market health. The minimal information-advantage model in Section 2 gives one mechanism for this pattern: liquid markets can rotate winners over time while still producing individual auctions in which one solver has a private advantage.

The paradox is useful because it rules out a common shortcut for monitoring intent systems. A protocol dashboard that only tracks the number of registered solvers or the long-run distribution of winners would classify blue-chip markets as relatively healthy. That classification is partly right: blue-chip cells have more long-run participation and less repeated concentration. It is also incomplete. The user does not trade against a long-run cell distribution; the user trades in one auction. If the realized winner has a private opportunity that other visible solvers do not share, the auction can be locally fragile even though the cell has many historical winners. Conversely, a long-tail cell can be persistently concentrated without always producing the largest relative winner-runner-up gap. The empirical lesson is therefore methodological: cell-level concentration, auction-level depth, and notional-normalized gaps should be monitored jointly; Figure 2 illustrates why none is a sufficient statistic for solver-market health.

Table 6 checks that the paradox is not a simple composition artifact. A blue-chip

■ **Table 4** HHI concentration after requiring minimum auction support per cell (all segments). Concentration persists after filtering sparse cells.

Min. auctions/cell	Cells	Med. HHI	HHI > 0.25	HHI = 1	Med. solvers
1	164,542	1.000	98.2%	66.7%	1
5	27,122	0.424	88.8%	5.7%	4
10	12,874	0.376	81.3%	2.7%	5
20	5,664	0.349	75.3%	1.8%	6
50	1,774	0.332	70.5%	1.4%	9

■ **Table 5** The blue-chip paradox. Blue-chip cells are less concentrated over time, yet blue-chip auctions have higher one-shot fragility than long-tail auctions. This shows that long-run solver diversity and auction-level replacement depth are distinct.

Segment	Frag > 0.5	Mean N^{comp}	Med. HHI	Med. solvers	Med. Gap ^{bps}
Blue-chip	32.1%	3.18	0.343	7	0.29
Long-tail	23.3%	2.58	1.000	1	2.56
Small auctions	63.1%	1.13	1.000	1	1.29
Large auctions	11.6%	3.07	1.000	1	1.28

indicator remains positive and significant after controlling for order size, month effects, effective solver count, cell-level HHI, and winner-solver fixed effects. The winner-solver fixed-effect specification is especially informative: the coefficient becomes larger, not smaller. Thus the pattern is within-solver as well as across-solver; the same solver is more likely to generate a high-fragility auction in blue-chip markets than in long-tail markets. This is consistent with the information-advantage model in Section 2: liquid markets have more visible participants over time, but they also create more opportunities for solver-specific RFQ, inventory, or internalization advantages that widen the one-shot winner-runner-up gap.

This interpretation also clarifies the role of solver identity. Winner rotation is not the same as close replacement. If different solvers take turns winning blue-chip auctions because different private opportunities arrive at different times, HHI can fall while one-shot fragility remains high. This is why the paper treats the blue-chip result as a paradox only relative to an overly simple market-health metric. Under the information-advantage model in Section 2, it is exactly the pattern one should expect in liquid markets with intermittent RFQ, inventory, or internalization advantages.

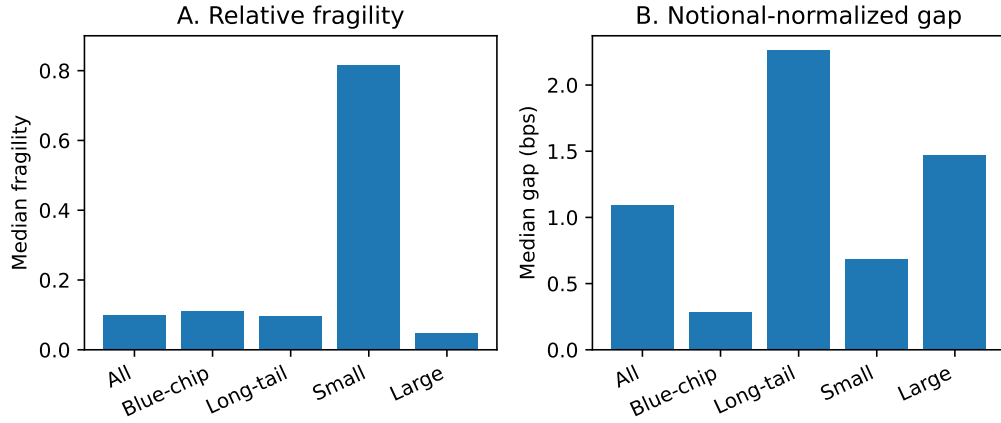
Economic scale, not causal identification.

Dollar replacement-gap proxies share score components with fragility, so they are not causal execution-quality estimates. Table 7 reports the economic scale on the USD-covered sane-volume sample. The score-implied visible replacement accounting gap totals \$5.13M over \$54.69B of covered notional, or 0.94 bps. Fragile auctions account for 37.1% of this covered-sample gap. These numbers are basis-point-scale rather than catastrophic, but still protocol-relevant because weak depth is concentrated in identifiable segments and can be monitored automatically.

Because contemporaneous gap regressions share score components with fragility, we do not use them as main evidence. Instead, Table 8 asks whether historical cell fragility predicts current accounting-gap proxies. The bivariate and controlled specifications show that lagged fragility contains predictive information without using the current auction's runner-up gap as the explanatory variable. Appendix Table 18 reports a cell fixed-effect robustness check;

■ **Table 6** Mechanism-consistency tests for the blue-chip paradox. Outcome is $\mathbf{1}\{\text{Fragility}_a > 0.5\}$. Entries report the coefficient on the blue-chip indicator from linear probability models; all reported coefficients are significant at $p < 0.001$. Controls are added to rule out simple compositional explanations.

Spec.	Controls	Blue-chip coef.	Alternative explanation addressed
(1)	none	+0.088***	descriptive gap
(2)	size fixed effects	+0.057***	not only order-size composition
(3)	month FE + N^{comp}	+0.072***	not only time trend or more solvers
(4)	+ cell HHI	+0.074***	not only concentration confounding
(5)	+ winner-solver FE	+0.143***	not driven by a single solver identity



■ **Figure 2** Relative and notional-normalized measures of weak solver competition. Relative fragility and Gap^{bps} do not collapse to one scalar notion of market health.

its sign changes because the within-cell question is different from the cross-cell comparison. We therefore read the table as predictive structure, not as a monotone causal relation.

Appendix D reports an approximate blue-chip router benchmark based on daily DEX reference prices [10]. It is directional rather than block-exact and should not be read as execution-quality identification. Still, it is an external sanity check in the expected direction: higher-fragility auctions compare more weakly against public DEX references, and the share of auctions beaten by the router rises from 52.3% in the lowest fragility bucket to 57.6% in the highest bucket. The effect is economically modest, which reinforces our interpretation of fragility as a visible replacement-depth and dependency-risk measure rather than a sufficient statistic for user execution harm.

Dynamics and persistence.

Figure 3 shows a descriptive improvement in visible replacement depth over the post-CIP-67 observation window. The monthly fragile-auction share falls from 28.1% in June to 20.0% in January, while mean competitive-solver count rises from 2.41 to 2.74. This makes the claim more nuanced. The solver market is not permanently frozen; participation and effective depth appear to improve over time. But the persistent tail remains large enough to justify measurement and protocol-side monitoring.

Table 9 gives a simple cell-month version of the same fact. Fragile share declines by 0.93 percentage points per month after cell demeaning, and higher mean effective solver count is

■ **Table 7** Economic scale of the visible replacement accounting-gap proxy (Jun. 2025–Jan. 2026, 244 days). The gap is a score-implied accounting difference, not an execution-loss or solver-profit estimate; denominator is the sane-volume USD-covered sample.

Quantity	Value
Main sample auctions	802,816
Sample days	244
Covered-volume auctions	318,469
Covered notional (USD)	\$54.69B
Total accounting-gap proxy (USD)	\$5.13M
Gap / covered notional	0.94 bps
Fragile-auction gap (fragility>0.5)	\$1.90M
Fragile-auction gap share	37.1%
Daily average gap	\$21,007
Annualized scale (extrapolated)	\$7.67M

■ **Table 8** Lagged cell fragility predicts current accounting-gap proxies. Predictor: cell-week mean fragility lagged one week. HC3 robust standard errors; *** $p < 0.001$.

Outcome	Specification	$\hat{\beta}_{\text{lag1}}$	SE	p	R^2	N
Score-diff gap (USD)	bivariate	−25.694***	(0.170)	< 0.001	0.021	727,784
Score-diff gap (USD)	+ controls	−20.077***	(0.256)	< 0.001	0.023	727,784
Reward-accounting gap (USD)	+ controls	−21.455***	(0.291)	< 0.001	0.017	617,402

Lagged predictors reduce contemporaneous score-component overlap. Appendix Table 18 reports a within-cell fixed-effect robustness check.

associated with lower fragile share even with cell and month fixed effects. These associations are consistent with market deepening, not a causal estimate of CIP-67.

The dynamic evidence is important for interpretation. A static measurement could be read as saying that solver competition is permanently broken. The eight-month panel suggests a more constructive view. Participation and effective depth improve over time, but the improvement is uneven and does not eliminate the fragile tail. That is the setting in which protocol-side monitoring is most useful: not as a claim that every auction requires intervention, but as a way to identify the cells and time periods where local discipline remains thin despite overall market learning.

Persistence is measurable at the cell level. Lagged cell fragility alone explains 40.6% of current variation and the controlled specification explains 63.9%; the full persistence table is in Appendix Table 17. This supports cell-level calibration: a benchmark based on historical cell behavior is using predictive information available before the current auction, not merely describing failures after the fact.

Solver identity and specialization.

A benign alternative hypothesis is narrow specialization: perhaps a solver dominates one pair because it is uniquely good at that pair. The identity analysis suggests a broader structural pattern. Many solvers are dominant in multiple pairs, and the top solver’s monthly share falls from 26.0% in June to 14.5% in January as the market deepens. The dominant winner persists from one week to the next in only 26.1% of cell-weeks; Appendix Table 19 reports the summary. That low persistence does not imply full competition: it means the identity of the dominant solver rotates, while many individual auctions still have weak visible replacement depth. This is exactly the point of using both identity persistence and auction-level fragility.

Taken together, these results give a coherent picture. Solver competition is improving,

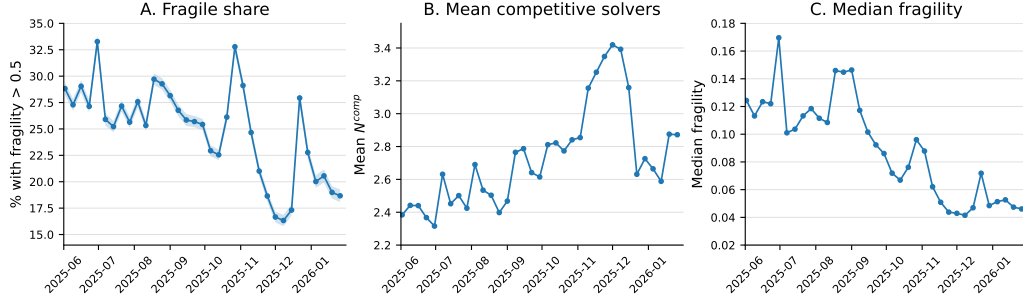


Figure 3 Descriptive post-CIP-67 evolution from June 2025 through January 2026. Panel A reports weekly fragile-auction shares with binomial 95% confidence bands; Panels B and C report mean competitive-solver count and median fragility. The figure is descriptive rather than a causal event study.

Table 9 Cell-month panel evidence of market deepening. Outcome: share of auctions with fragility > 0.5 per cell-month. Both specifications use within-cell demeaning. HC3 robust standard errors. This is descriptive and does not identify a causal CIP-67 effect.

Specification	Predictor	Coef.	SE	p	R^2
month index + cell FE	Month index	-0.0093***	(0.0005)	< 0.001	0.020
$N^{comp} + \text{cell} + \text{month FE}$	Mean N^{comp}	-0.0356***	(0.0011)	< 0.001	0.048

accounting-gap proxies are basis-point-scale but protocol-relevant, concentration is local and robust to support filters, blue-chip markets reveal a difference between long-run diversity and one-shot replacement depth, and solver advantages look general rather than purely pair-specific.

5 Shadow-Reserve Benchmarks

The measurement results motivate a benchmark, not a replacement for solver competition. A shadow reserve is a protocol-maintained score benchmark for a market cell. It can be used to audit low-quality auctions, adjust rewards, or trigger a fallback when an executable route is available. The mechanism studied here is a quality floor and reward-adjustment primitive. It is not a full fallback-routing system and it is not designed to trigger exactly on high relative fragility.

For a market cell c and time t , let $r_{c,t}$ be the reserve. The reserve is built from delayed historical samples:

$$\mathcal{H}_{c,t} = \{z_{c,u} : t - L \leq u \leq t - H, N_u^{\text{comp}} \geq k, n_{c,u} \geq m\}.$$

Following robust benchmark practice [17, 18], the provisional benchmark is a winsorized lower-tail quantile, $\hat{r}_{c,t} = Q_\tau(\mathcal{H}_{c,t})$. The update is smoothed and rate-limited:

$$r_{c,t+1} = \max\{(1 - \delta_\downarrow)r_{c,t}, \min[(1 + \delta_\uparrow)r_{c,t}, (1 - \eta)r_{c,t} + \eta\hat{r}_{c,t}]\}.$$

The delay prevents immediate feedback, the support and competition filters reduce sparse-cell noise, the robust quantile limits lower-tail manipulation, and the downward rate limit makes reserve depression slow and costly.

The baseline values are $L = 200$, $H = 10$, $k = 2$, $\tau = 0.25$, $\eta = 0.10$, $\delta_\downarrow = 0.05$, and $\delta_\uparrow = 0.10$; Appendix Table 22 lists the full parameter table. Appendix Table 23 varies the

key parameters. Increasing the quantile from 0.10 to 0.40 raises activation from 15.4% to 29.1% and increases accounting-gap compression, while changing the downward rate limit has smaller short-run effects. This is not an optimization exercise; it exposes the deployment tradeoff between a stricter benchmark and a lower-intervention benchmark.

Quality floor rather than fragility target.

The reserve triggers when the winner’s absolute score is below a historical benchmark. Fragility is relative, $1 - \text{Score}_2^{\text{other}} / \text{Score}_1$. A high-fragility auction may have a strong absolute winner and a weak runner-up, so the quality floor need not bind. Appendix Table 24 reports the decile diagnostic: activation is lower in the highest-fragility deciles. A fragility-targeted reserve would instead monitor $\text{Score}_2^{\text{other}} / \text{Score}_1$ or the full visible score-depth curve.

This limitation is useful. It turns a possible criticism into a design distinction: quality floors protect against all-bids-weak auctions, while depth-targeted reserves protect against relative winner dominance. The present paper evaluates the former and diagnoses the latter as a complementary design primitive.

Depth-targeted diagnostic.

The quality-floor trigger and a depth-targeted trigger flag different failure modes. The quality floor asks whether the winner’s absolute score falls below a historical benchmark. A depth trigger instead asks whether the visible depth ratio $\text{Score}_{2,a}^{\text{other}} / \text{Score}_{1,a}$ is low. Table 10 compares the baseline quality floor to three depth diagnostics. The quality floor flags only 14.4% of the highest-fragility decile, while the simple triggers $\text{Depth} < 0.50$ and $\text{Depth} < 0.25$ flag 100% of that decile and overlap with the quality floor in only about 16% and 15% of triggered auctions. Thus all-bids-weak auctions and winner-dominance auctions are distinct protocol-design problems. The baseline reserve is an absolute quality floor; a production fragility-targeted reserve would combine it with a depth monitor.

■ **Table 10** Depth-targeted reserve diagnostic. The baseline quality-floor trigger ($\text{Score}_1 < r_{c,t}$) and depth triggers flag different failure modes. “Top decile activation” is the share of the highest-fragility decile flagged by the rule; “Overlap” is the fraction of triggered auctions also flagged by the quality floor.

Trigger	Activation	Top decile act.	Overlap w/ QF	Med. gap proxy
Quality floor (baseline)	21.0%	14.4%	100.0%	\$1.02
Depth < 0.50 (fragility > 0.5)	24.6%	100.0%	16.1%	\$4.06
Depth < 0.25 (fragility > 0.75)	19.7%	100.0%	15.3%	\$6.21
Cell historical depth Q25	19.7%	35.9%	30.7%	\$11.22

The diagnostic should be read as a design observation rather than a new theorem. The baseline quality floor asks, “is the winning score unusually low for this cell?” The depth trigger asks, “is the winner insufficiently disciplined by a visible alternative?” These questions are empirically non-substitutable. A protocol can therefore combine them without treating either as a universal execution-quality test. For example, a soft reward adjustment could use the quality floor to flag all-bids-weak auctions, while a depth monitor could trigger additional auditing, solver onboarding incentives, or requirements for a stronger fallback route in cells where winner dominance is recurrent.

The two-trigger view also explains why the quality floor should not be judged by its activation rate in the highest-fragility decile alone. A high-fragility auction can be a

high-quality execution with a private winner advantage; the appropriate response may be monitoring or disclosure rather than rejection. Conversely, an all-bids-weak auction can have low relative fragility because the winner and runner-up are both poor. Separating these cases avoids the common mistake of using one reserve rule to represent all forms of weak competition.

Deployment tradeoffs.

The same benchmark can be deployed at different strengths. A soft reserve is safest for fill rate and works as a reward or audit primitive. A non-executable hard reserve gives a stronger quality-floor interpretation but risks rejecting trades without a credible fallback. An executable hard reserve requires contemporaneous router or protocol-owned fallback data and therefore has the strongest execution-quality interpretation but the highest implementation burden. Appendix Table 25 summarizes this tradeoff matrix.

The empirical replay results should therefore be read under the soft/accounting interpretation. They quantify how much visible replacement accounting gap a benchmark can compress under stylized response scenarios. They do not claim that a deployed hard reserve would have the same execution-quality effect without modeling fill rates, gas, slippage, and solver participation.

6 Calibration and Manipulation Diagnostics for Endogenous Benchmarks

The basic hard-reserve properties are design facts rather than deep theorems. If an executable reserve belongs to the feasible set, maximizing score cannot choose a worse feasible outcome. If a bid is below a hard reserve, it cannot win under the hard-threshold interpretation. If real competition already clears the reserve, the reserve does not change the winner. These facts justify the terminology, but the central deployment issue is endogenous benchmark manipulation.

Any non-trivial benchmark computed from historical bids can be manipulated if a coalition controls all benchmark-eligible samples; robust quantile estimators provide only bounded-contamination protection [16, 17, 18]. The design therefore filters benchmark updates by competition and support, delays the update window, winsorizes samples, uses a robust quantile, and rate-limits downward movement. Let F_0 be the distribution of uncontaminated benchmark samples, let a coalition contribute at most an α fraction of samples, and let the observed distribution be $F = (1 - \alpha)F_0 + \alpha G$. If the coalition places all contaminated mass at the lower tail and $\tau > \alpha$, then

$$Q_\tau(F) \geq Q_{(\tau-\alpha)/(1-\alpha)}(F_0).$$

Thus contamination can move the effective quantile downward, but only to the adjusted uncontaminated quantile. The downward rate limit adds an intertemporal cost: lowering a reserve by a fraction Δ requires at least approximately $\Delta/\delta_\downarrow$ successful downward-update periods.

Protocol design corollary.

The relevant empirical diagnostic is not the number of active solvers alone, but the largest share of benchmark-eligible samples controlled by a solver or coalition. With the baseline $\tau = 0.25$, a cell in which the largest eligible-sample share is below 25% has a robust-quantile

buffer against one-solver reserve depression. A cell in which a solver exceeds this share, or a cell with too little support, should not maintain an independent hard benchmark. It should inherit from a parent cell, use a soft reserve, or be treated as audit-only. Appendix Table 26 summarizes this rule.

Empirical vulnerability diagnostic.

Table 11 shows that the robust-quantile condition is operationally stringent in local solver markets. Across cell-windows, 99.6% are vulnerable under the unweighted diagnostic and 97.1% are vulnerable under auction-weighting; even blue-chip cells are vulnerable in 95.3% of auction-weighted exposure. These numbers are high partly because many local cells are sparse, but that is exactly the deployment lesson: a standalone hard reserve at the finest cell level would often be controlled by too few eligible observations. Parent-cell inheritance and audit-only soft reserves are therefore not theoretical niceties; they are necessary deployment rules for most cells.

Table 11 Reserve benchmark vulnerability diagnostic. A cell-window is vulnerable if the maximum eligible-sample share of any single solver is at least the reserve quantile $\tau = 0.25$. Auction-weighted shares account for cell size.

Segment	Cell-windows	Vuln. (unwtd)	Vuln. (auction-wtd)	Sparse (< 5)	Med. max share
all	91,390	99.6%	97.1%	86.4%	1.000
blue-chip	1,316	96.3%	95.3%	22.1%	0.500
long-tail	90,074	99.6%	97.8%	87.4%	1.000
small	9,740	99.9%	98.9%	93.9%	1.000
large	31,597	99.4%	94.6%	85.6%	1.000

The high vulnerability rates are not evidence that reserves fail; they show that most local cells should inherit parent-cell benchmarks or use audit-only/soft reserves rather than independent hard reserves.

The diagnostic is not evidence that solvers are manipulating reserves. It is a deployment stress test. A cell-window can be vulnerable simply because too few eligible observations remain after applying support and competition filters, or because one solver frequently supplies the only benchmark-eligible observations. In such cells, an independent hard reserve would inherit the market's thinness and could be too easy to move. The recommended response is not to abandon benchmarks, but to choose the benchmark granularity to match the data. Parent-cell inheritance, audit-only soft reserves, and coarser benchmarks are ways to preserve monitoring value without pretending that a sparse local cell supports a hard, manipulation-resistant reserve.

This theoretical framing matches the empirical interpretation. The reserve is a manipulation-aware benchmark with transparent tradeoffs. It is not an optimal mechanism for solver entry, nor an execution-quality guarantee without executable fallback routes. Its value is that the protocol can choose and audit a point on a clear robustness-responsiveness tradeoff curve.

7 Accounting Replays

We evaluate the reserve through accounting replays. These are not equilibrium deployment simulations. They do not model fallback routing, latency, gas, or long-run solver entry and exit. They ask a narrower question: if the protocol used the historical benchmark as a reward or quality-floor primitive, how often would it bind and what visible replacement accounting gap would it compress?

Algorithm 2 Behavior-aware threshold-response replay

Require: Auction score pair $(Score_1, Score_2^{other})$, reserve r_a , margin ϵ , cost buffer κ

- 1: **if** $Score_1 \geq r_a + \kappa$ **then**
- 2: Winner remains feasible; adjusted bid clears $\max\{Score_1, r_a + \epsilon\}$.
- 3: **else if** $Score_2^{other} \geq r_a + \kappa$ **then**
- 4: Winner exits; best non-winning solver clears $\max\{Score_2^{other}, r_a + \epsilon\}$.
- 5: **else**
- 6: No visible solver clears reserve; mark reserve/fallback-needed.
- 7: **end if**
- 8: Record accounting-gap change relative to historical accounting gap.

487 Mechanical replay leaves historical bids unchanged. Behavior-aware replay is an optimistic
 488 threshold-response sensitivity scenario: if a solver can clear $r_a + \kappa$, its bid is adjusted around
 489 the reserve threshold; otherwise it may exit and the next visible solver can replace it.
 490 Adversarial replay stress-tests the reserve lower-bound property when solvers bid near the
 491 reserve. These scenarios form a diagnostic bracket, not a mathematical bound on real
 492 deployment effects.

493 The adversarial stress test should be read as a deployment warning rather than a proposed
 494 hard-reserve policy. In near-threshold strategic conditions the reserve can activate much
 495 more frequently, so a non-executable hard reserve would carry substantial fill-risk unless
 496 a credible fallback route exists. This is why the deployment recommendation in Section 5
 497 defaults to soft/audit use for sparse cells and treats executable hard reserves as appropriate
 498 only when contemporaneous fallback routing is available.

499 Appendix Table 27 reports the activated/non-activated decomposition. It explains a
 500 pattern that otherwise looks puzzling. In mechanical replay, only activated auctions change,
 501 so non-activated auctions have zero accounting-gap change. In behavior-aware replay, non-
 502 activated auctions can also compress accounting gap because the reserve changes the threshold
 503 around which solvers shade bids. This is why the non-activated behavior-aware mean is
 504 negative. We do not interpret this as an equilibrium prediction; it is a sensitivity check
 505 showing that accounting-gap compression is not entirely driven by changing the winner in
 506 activated auctions.

507 The sensitivity results in Table 23 show that the accounting effect is not a single cherry-
 508 picked parameter setting. Higher quantiles activate more often and compress larger accounting
 509 gaps; lower quantiles are more conservative. Smaller downward rate limits make the bench-
 510 mark more manipulation-resistant but slower to respond. These tradeoffs are more useful for
 511 protocol design than a single point estimate.

512 **8 Discussion and Conclusion**

513 The paper measures visible replacement depth, not every possible form of market power.
 514 CoW solver scores are meaningful protocol-side accounting values, and the corrected parser
 515 constructs the exact best non-winning solver reference score for the observed single-winner
 516 setting. But the data do not reveal all private RFQ relationships, inventories, communications,
 517 or globally optimal fallback routes. Dollar replacement-gap proxies translate the visible
 518 score difference into economic magnitude; they are not causal execution-quality estimates,
 519 solver-profit estimates, or total user-surplus losses. Table 12 summarizes these interpretation
 520 boundaries.

■ **Table 12** How to interpret the main empirical objects. The paper treats these quantities as observable diagnostics of solver competition, not as direct execution-loss, profit, or collusion estimates.

Quantity	What it supports	What it does not support
Visible fragility	Auction-level replacement depth	User harm, collusion, global optimality
Accounting-gap proxy	Score-implied dollar scale	Solver profit or causal execution loss
Router sanity check	Directional external validation	Block-exact counterfactual execution quality
Cell-level HHI	Persistent local concentration	One-shot replacement depth
Quality-floor trigger	All-bids-weak monitoring	Winner-dominance monitoring
Depth trigger	Winner-dominance monitoring	Absolute execution-quality guarantee
Accounting replay	Audit/reward sensitivity	Equilibrium deployment effect

External validity.

Some lessons are broader than CoW. A high registered solver or filler count is not enough; competition must be measured locally at the auction and market-cell level. Long-run diversity and one-shot replacement depth are different objects, as the blue-chip paradox illustrates. Small and thin auctions are vulnerable to weak visible replacement because fixed search and optimization costs matter. Other intent systems such as UniswapX, ERC-7683-style cross-chain intent networks, or 1inch Fusion may not expose losing bids with the same richness [20, 2, 1], but the same measurement question remains: does the accepted filler face close local discipline? What is CoW-specific is the exact dollar scale, the reference-score accounting construction, and the reserve calibration parameters. Appendix Table 29 summarizes the observability differences across CoW, UniswapX, and 1inch Fusion.

Design implications.

First, protocols should monitor both auction-level fragility and cell-level concentration. Second, reserves should be matched to deployment context. Soft reserves are appropriate where fill risk dominates; executable hard reserves require contemporaneous fallback routing. Third, if a protocol wants to target relative winner dominance, it should condition on visible depth ratios, not just absolute score shortfall. Fourth, benchmark manipulation is a cell-level problem: support-poor or highly dominated cells should inherit benchmarks from coarser parent cells or use audit-only reserves.

Several limitations remain. USD notional coverage is incomplete, although missingness diagnostics show that uncovered auctions are not more fragile on the median. The reserve replays are accounting exercises, not equilibrium predictions. HHI depends on market-cell definitions, though support filtering and robustness checks mitigate sparse-cell concerns. Finally, the appendix reports a directional blue-chip router benchmark using daily DEX reference prices; a stronger execution-quality validation would require contemporaneous or block-specific router reconstruction. Pre-CIP-67 causal evaluation also remains a valuable extension, but it requires comparable pre-transition records.

Intent-based execution creates a new market-design object: solver competition itself. This paper provides a large-scale measurement of its visible depth, accounting-gap scale, persistence, and benchmark-based mitigation. The main lesson is not that solver markets are broken. It is that solver competition is local, heterogeneous, and auditable. Protocols should therefore measure replacement depth and design safeguards around the market cells where visible competition is thin.

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611 A Additional Proof Details

612 The contamination bound in Section 6 follows from the definition of quantiles under Huber-
613 style contamination. If a coalition of mass α places all contaminated observations below the
614 support of the clean distribution F_0 , then the τ quantile of the contaminated distribution
615 corresponds to the $(\tau - \alpha)/(1 - \alpha)$ quantile of the uncontaminated part. Any other contami-
616 nation distribution can only raise the contaminated τ quantile. The downward rate limit
617 gives the intertemporal manipulation bound because after T downward periods the fastest
618 possible decline is approximately $(1 - \delta_{\downarrow})^T$.

619 B Additional Measurement and Robustness Tables

620 This appendix collects measurement checks that support Sections 3 and 4 without inter-
621 rupting the main empirical narrative. The first group documents USD missingness, raw
622 local concentration, decile-scale outcomes, contemporaneous scale-translation regressions,
623 persistence, and solver identity. Tables 15, 16, and 20 provide the decile-scale, regression,
624 and compact robustness checks.

■ **Table 13** USD coverage diagnostics. Full-sample fragility and HHI do not rely on USD coverage; gap proxies and Gap^{bps} use the covered subsample.

Sample	N	Share	Med. frag.	Frag > 0.5	Mean N^{comp}	Blue-chip share
full	802,816	100.0%	0.086	25.3%	2.72	22.7%
USD-covered	493,653	61.5%	0.084	26.6%	2.86	35.8%
USD-uncovered	309,163	38.5%	0.089	23.2%	2.49	1.8%

■ **Table 14** Local concentration by segment. HHI is computed over market cells.

Segment	Cells	Med. HHI	HHI > 0.25	HHI = 1	Med. auctions	Med. solvers
all	164,542	1.000	98.2%	66.7%	1	1
blue-chip	1,404	0.343	71.4%	4.5%	39	7
long-tail	163,138	1.000	98.4%	67.3%	1	1
small	39,677	1.000	99.2%	70.6%	1	1
large	45,682	1.000	97.3%	65.1%	1	1

■ **Table 15** Dollar replacement-gap proxy by fragility decile.

Decile	Med. fragility	Med. gap proxy (USD)	Med. gap (bps)	<i>N</i>
D1	0.001	\$0.24	0.07	68,084
D2	0.005	\$0.68	0.32	68,084
D3	0.012	\$0.93	0.69	68,084
D4	0.023	\$1.06	1.22	68,084
D5	0.041	\$1.05	1.89	68,084
D6	0.070	\$1.10	2.79	68,083
D7	0.120	\$1.15	2.99	68,084
D8	0.216	\$1.43	2.85	68,084
D9	0.410	\$2.13	4.11	68,084
D10	0.800	\$4.33	5.78	68,084

■ **Table 16** Economic-scale regressions. USD replacement-gap regressions are interpreted as scale translations of visible replacement gaps, not causal estimates.

Outcome	FE	Coef.	SE	<i>p</i>	<i>R</i> ²	<i>N</i>
gap proxy (USD)	none	65.31	0.54	<0.001	0.075	680,839
gap proxy (USD)	pair+chain	69.26	0.57	<0.001	0.077	680,839
Gap (bps)	none	-653.95	16.10	<0.001	0.106	493,653
Gap (bps)	pair+chain	-99.77	9.87	<0.001	0.390	493,653

■ **Table 17** Fragility persistence. Lagged cell fragility strongly predicts current fragility, motivating cell-level calibration.

Specification	<i>N</i>	<i>R</i> ²	Lag-1 coef.	SE	<i>p</i>
Lag-1 only	727,784	0.406	0.799	0.0013	<0.001
Controls	727,784	0.639	0.295	0.0017	<0.001

■ **Table 18** Lagged predictor robustness, including the within-cell fixed-effect specification. The cell-FE row answers a different within-cell question and is not interpreted as a monotone causal effect.

Outcome	Specification	$\hat{\beta}_{\text{lag1}}$	SE	<i>p</i>	<i>R</i> ²	<i>N</i>
Score-diff gap (USD)	bivariate	-25.694***	(0.170)	< 0.001	0.021	727,784
Score-diff gap (USD)	+ controls	-20.077***	(0.256)	< 0.001	0.023	727,784
Score-diff gap (USD)	+ cell FE	0.844**	(0.305)	0.006	0.006	727,784
Reward-accounting gap (USD)	+ controls	-21.455***	(0.291)	< 0.001	0.017	617,402

Across cells, historically fragile cells often correspond to thin or small markets with lower dollar gaps; within a cell, weeks with above-usual lagged fragility have a small positive association with the current gap. Both specifications are descriptive.

■ **Table 19** Solver identity concentration and persistence. Winner identities rotate over time, but repeated dominance by large solvers across many pairs suggests structural capability advantages rather than narrow token-pair specialization.

Statistic	Value
Unique solvers in sample	34
Solvers dominant in at least five pairs	26
Top-1 monthly winner share, June 2025	26.0%
Top-1 monthly winner share, January 2026	14.5%
Dominant winner persists to next week	26.1%
Blue-chip top-1 winner share	31.3%
Long-tail top-1 winner share	23.7%

■ **Table 20** Compact robustness summary. “Yes” indicates that the qualitative finding remains in the specified check.

Specification	Thick tail	Blue-chip paradox	HHI persists	Gap scale	Reserve effect
Baseline full sample	Yes	Yes	Yes	Yes	Yes
USD-covered only	Yes	Yes	n/a	Yes	n/a
High-support cells (≥ 50)	n/a	n/a	Yes	n/a	n/a
Corrected parser	Yes	Yes	Yes	Yes	Yes
Reserve sensitivity	n/a	n/a	n/a	n/a	Stable
Jun.–Sep. subsample	Yes	Yes	Yes	Yes	Yes

625 **Why the corrected reference-score construction matters.**

626 The corrected parser changes the measurement in a conservative direction. The earlier solution-
627 level runner-up construction could understate fragility whenever the winner submitted
628 multiple high-scoring candidate solutions. In that case, the second-highest solution overall
629 was not a credible outside option for the user, because it came from the same winning solver.
630 The proper reference is the best score attainable after removing the winning solver. This
631 distinction matters operationally: a protocol that wants to discipline a winner must compare
632 it to a different solver, not to the same solver’s second candidate. The validation table
633 shows that such cases occur in a non-trivial share of auctions. The measurement framework
634 therefore uses the best non-winning solver score throughout the main analysis.

635 **Toy example.**

636 Table 21 illustrates why the parser groups by solver identity. A winning solver’s second
637 candidate solution is not a credible outside option after removing the winner; the correct
638 reference is the best score submitted by another solver.

■ **Table 21** Solution-level runner-up toy example. The winner’s second solution is not a credible outside option because it comes from the same solver. The correct reference score removes the winning solver before taking the maximum remaining score.

Solver	Solution	Score	Role
A	A1	100	winning solution
A	A2	99	same-solver runner-up, not outside option
B	B1	60	best non-winning solver score

639 Why USD coverage does not drive the main measurement.

640 The primary objects of the paper—fragility, effective solver counts, and cell-level HHI—are
 641 computed before any USD enrichment. These quantities are available whenever the auction
 642 has valid positive solver scores. USD enrichment is used only to express scale in dollars
 643 and to compute notional-normalized basis-point gaps. This separation is important because
 644 token price coverage is uneven across long-tail assets. The missingness diagnostics show that
 645 the covered sample contains a larger blue-chip share and a higher mean effective solver count.
 646 We therefore avoid using USD-denominated estimates to make full-population claims. When
 647 reporting dollar gap proxies and bps gap, the paper explicitly conditions on the covered
 648 sample and reports the corresponding denominator.

649 Why market cells are local.

650 Market cells are not meant to be universal definitions of a market. They are measurement
 651 units for the economic environment in which a solver is actually competing. Solvers differ in
 652 token inventory, RFQ access, gas strategies, private routing relationships, and risk limits.
 653 These differences are naturally local to token pair, order size, chain, and time. At the
 654 same time, a cell definition that is too fine can create sparse cells and inflate HHI. The
 655 paper therefore reports both raw local concentration and support-filtered concentration. The
 656 support-filtered results are the main defense against the sparse-cell artifact, while the raw
 657 cells show how often the user-facing auction is effectively local and thin.

658 C Additional Reserve and Replay Tables

659 This appendix collects reserve-parameter, sensitivity, targeting, deployment, and replay
 660 tables that support Sections 5–7. The main text uses the depth-targeted diagnostic and
 661 benchmark-vulnerability diagnostic as the central design evidence. The tables below expose
 662 the parameter values and replay diagnostics behind that interpretation.

■ **Table 22** Baseline shadow-reserve parameters.

Parameter	Value	Rationale
Lookback L	200 auctions	historical benchmark window
Delay H	10 auctions	prevents immediate feedback manipulation
Competitive threshold k	2	benchmark samples require competition
Quantile τ	0.25	robust lower-tail benchmark
Smoothing η	0.10	gradual reserve update
Downward limit $\delta_{\downarrow}$	0.05	manipulation resistance
Upward limit $\delta_{\uparrow}$	0.10	bounded upward response
Winsorization	5%/95%	outlier control
Warm-up	20%	initialize reserve states

■ **Table 23** Reserve sensitivity to benchmark parameters.

Setting	Activation	Overall Δ gap	Active Δ gap
baseline	22.6%	\$-1.83	\$-8.10
tau=0.10	15.4%	\$-1.16	\$-7.50
tau=0.20	20.3%	\$-1.61	\$-7.92
tau=0.40	29.1%	\$-2.67	\$-9.17
delta_down=0.02	23.2%	\$-1.99	\$-8.57
delta_down=0.10	22.5%	\$-1.77	\$-7.88
eta=0.05	22.9%	\$-1.88	\$-8.21
eta=0.20	22.5%	\$-1.82	\$-8.10

■ **Table 24** Shadow-reserve activation rate by fragility decile. The reserve activates less in the highest-fragility auctions, confirming that the mechanism is a quality floor rather than a fragility-targeted trigger.

Decile	Mean fragility	Activation	Winner change	N
D1	0.001	19.0%	19.0%	64,226
D2	0.006	23.0%	23.0%	64,225
D3	0.015	22.8%	22.8%	64,225
D4	0.031	23.8%	23.8%	64,225
D5	0.059	23.6%	23.6%	64,226
D6	0.112	23.2%	23.2%	64,225
D7	0.228	23.5%	23.5%	64,225
D8	0.500	20.6%	20.6%	64,225
D9/D10	0.972	15.4%	15.4%	128,451

■ **Table 25** Deployment tradeoffs for shadow-reserve benchmarks. The paper’s empirical replays correspond to the soft/accounting interpretation; stronger modes require additional fallback infrastructure.

Mode	Settlement impact	Fallback?	Main risk	Recommended use
Soft reserve	reward/audit only	no	weak execution-quality guarantee	sparse or long-tail cells
Non-exec hard	reject below benchmark	no	fill-rate loss	liquid cells with solver depth
Exec hard	route to fallback	yes	gas/slippage/latency	blue-chip pairs with live router data

■ **Table 26** Protocol design interpretation of the robust-quantile condition. The relevant diagnostic is the largest eligible-sample share, not merely the number of active solvers.

Condition	Reserve implication	Practical action
Max eligible share $< \tau$	single-coalition depression limited	maintain cell-level reserve
Max eligible share $\geq \tau$	benchmark vulnerable to dominant solver	inherit parent-cell benchmark
Support too sparse	robust quantile unreliable	audit-only or soft reserve
Executable fallback available	stronger user-surplus interpretation	consider executable hard reserve

■ **Table 27** Counterfactual replay decomposition on the eight-month sample. Behavior-aware replay is a threshold-response sensitivity scenario, not an equilibrium estimate. Non-activated auctions can still compress accounting-gap proxy because the reserve changes the bid-adjustment threshold.

Replay	N	Activation	Overall Δ Gap	Active Δ Gap	Non-active Δ Gap	Thm. 1
Mechanical	642,297	21.0%	\$-1.57	\$-7.47	\$0.00	—
Behavior-aware	642,297	21.0%	\$-5.81	\$-8.32	\$-5.09	—
Adversarial	642,253	66.2%	—	—	—	100%

Why the reserve is not claimed to be optimal.

The reserve benchmark is intentionally modest. A full optimal mechanism would require a model of solver entry, private inventory, off-chain RFQ agreements, strategic withdrawal, execution costs, and fallback routing quality. Such a model is outside the scope of this measurement paper and would require assumptions not observed in the data. The reserve instead provides a transparent benchmark with interpretable parameters. The sensitivity table exposes how activation and accounting-gap compression move with those parameters. This is useful for protocol design because a DAO or protocol operator can choose a point on the tradeoff curve rather than accept an opaque intervention.

D External Benchmark and Cross-Protocol Scope

The main paper treats public-router validation as complementary evidence rather than as a full execution-quality counterfactual. Table 28 reports the directional blue-chip router benchmark against daily DEX reference prices. It is useful as an external sanity check, but it is not block-exact and therefore should not be interpreted as a causal execution-quality estimate. Table 29 summarizes which parts of the measurement framework travel beyond CoW and which rely on CoW-specific visibility into losing solver submissions.

Table 28 Approximate blue-chip router benchmark. CoW execution is compared with a daily DEX reference from DeFiLlama; positive improvement means CoW is better. This is directional validation only, not a block-exact execution-quality counterfactual; the benchmark is intentionally narrow.

Fragility bucket	N	Med. impr. (bps)	Router-better	Mean frag.
0–0.1	195,780	-1.5	52.3%	0.025
0.1–0.3	45,036	-1.5	53.3%	0.179
0.3–0.5	20,171	-1.5	53.9%	0.392
0.5–0.75	17,710	-1.9	55.1%	0.620
0.75–1.0	63,570	-2.3	57.6%	0.964
High-fragility coefficient: 6.14***, $p < 0.001$, $N = 332,845$				

Table 29 Cross-protocol observability of replacement depth. CoW exposes losing solver scores, so best-non-winner fragility is directly auditable. Other intent systems may require weaker proxies unless losing filler quotes or off-chain logs are available.

Protocol	Execution object	Losing bids visible?	Best available depth proxy
CoW Protocol	solver score competition	yes	best-non-winner fragility, N^{comp} , HHI
UniswapX	filler / Dutch auction execution	generally no	filler concentration, fill timing, execution vs. limit
1inch Fusion	resolver-based intent execution	generally no	resolver concentration, fill latency, execution outcome

What an external benchmark would add.

A block-specific public-router or live-quote validation would answer a different question: whether high visible fragility is associated with weaker improvement over public liquidity. That would move the analysis closer to execution-quality validation. The current work establishes the visible structure of solver competition and provides a reproducible benchmark methodology. External execution-quality validation is the natural next step, but it should be performed with contemporaneous quotes or block-specific historical routing to avoid using aggregate or today’s liquidity to evaluate past auctions.

E Artifact and Reproducibility

Raw data are CoW Protocol solver-competition JSON records in the post-CIP-67 format. The parser retains non-filtered positive-score solutions, identifies the winning solver, and computes the reference score as the best non-winning solver score. USD enrichment uses ETH/USD prices for score conversion and sell-token prices for notional coverage. The released artifact contains collection scripts, the corrected parser, USD enrichment scripts, metric construction, HHI robustness, reserve replay, validation scripts, table-generation scripts, and aggregated tables. Large per-auction parquet files exceed ordinary GitHub rendering limits and are represented in the paper by deterministic CSV summaries and generated LaTeX tables. Table 30 maps paper objects to scripts, inputs, and outputs.

Table 30 Artifact script-to-output map. Objects are named by stable paper artifact rather than by table number, so the map remains valid if floats are reordered. Large per-auction parquet files are stored separately from the lightweight paper artifact.

Paper object	Script	Input	Output
Sample construction	sample_flow.py	auctions_full_usd.parquet	sample_flow.csv
Parser audit	reference_score_validation.py	auctions_full.parquet	reference_score_validation.csv
USD missingness diagnostics	usd_missingness.py	auctions_full_usd.parquet	usd_missingness.csv
Main fragility summary	fragility.py	auctions_full_usd.parquet	fragility_summary_usd.csv
HHI concentration and support filtering	hhi_robustness.py	auctions_full_usd.parquet	HHI summary CSVs
Economic-scale accounting gap	economic_scale.py	auctions_full_usd.parquet	economic_scale.csv
Lagged predictor	lagged_predictor.py	auctions_full_usd.parquet	lagged_predictor.csv
Blue-chip consistency check	bluechip_consistency.py	auctions_full_usd.parquet	bluechip_consistency.csv
Depth-trigger diagnostic	depth_targeted_reserve_diagnostic.py	metrics + replay output	depth_targeted_reserve.csv
Benchmark vulnerability diagnostic	vulnerability_diagnostic.py	reserve windows	vulnerability_diagnostic.csv
Router sanity check	router_benchmark.py	blue-chip executions + prices	router_benchmark.csv
Reserve replay summaries	mechanical_replay.py	metrics + params	replay summary CSVs

Artifact checklist.

A complete replication of the paper requires: raw solver-competition records or the processed parquet files; the corrected parser that computes the best non-winning solver score; USD enrichment scripts and token-decimal mappings; metric scripts for fragility, effective solver counts, HHI, and USD-normalized gaps; scripts generating the sample-flow and missingness diagnostics; shadow-reserve replay scripts; and the table-generation script. The repository stores large intermediate files separately from the lightweight CSV summaries used in the paper because some per-auction files exceed ordinary GitHub rendering limits.

F Additional Design Alternatives

The quality-floor reserve studied in the paper is only one point in a larger design space. A fragility-targeted reserve would monitor the ratio $Score_2^{other}/Score_1$ directly and could trigger when the visible depth ratio falls below a historical cell-level benchmark. Such a mechanism would target relative winner dominance more directly than the quality-floor reserve, but it would have different costs: it could flag auctions with a high-quality winner merely because the second-best solver is far behind. A protocol might therefore combine both mechanisms. The absolute reserve protects users from low-quality execution relative to history, while a depth reserve flags auctions where the winner is insufficiently disciplined by visible alternatives.

A second extension is entry-aware calibration. If a reserve is too strict, marginal solvers may avoid markets in which they expect to lose against the benchmark. If it is too weak, it becomes a passive monitor. One possible implementation is a soft reserve that reduces rewards smoothly rather than rejecting settlements. For example, a solver could receive a reward discount proportional to the shortfall below the benchmark, capped to avoid

destabilizing participation. Such a soft variant would preserve fill probability while still making weak-competition accounting gaps visible to the protocol.

A third extension is oracle-assisted fallback. When a public route or protocol-owned fallback solver is available, a reserve can become executable rather than purely accounting-based. The execution-quality interpretation is stronger in that case, but only if the fallback is contemporaneous, gas-aware, and slippage-aware. The present paper avoids claiming such a guarantee because the historical dataset does not contain block-specific public-router counterfactuals. This limitation is also a practical design warning: a reserve without a credible fallback is best treated as a benchmark, not as a hard promise of user execution quality.